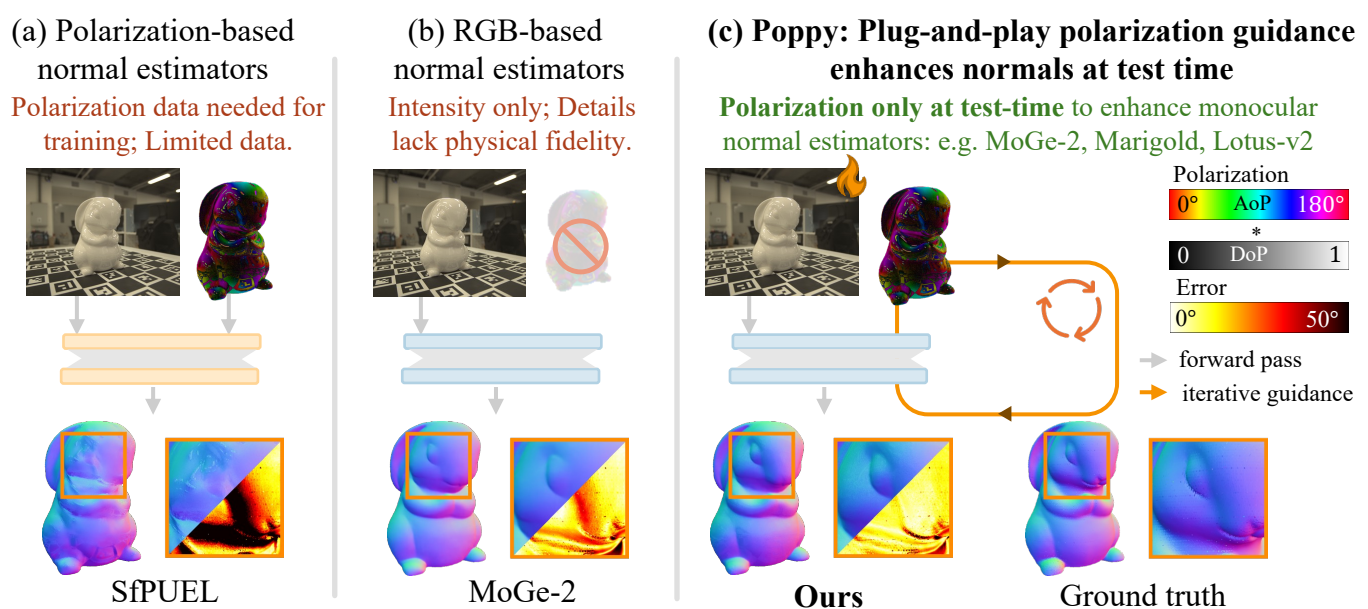
(a) Polarization-based normal estimators
Polarization data needed for training; Limited data.
(b) RGB-based normal estimators
Intensity only; Details lack physical fidelity.
(c) Poppy: Plug-and-play polarization guidance enhances normals at test time
Polarization only at test-time to enhance monocular normal estimators: e.g. MoGe-2, Marigold, Lotus-v2
Polarization
0° AoP 180°
*
0 DoP 1
Error
0° 50°
forward pass
iterative guidance
SfPUEL
MoGe-2
Ours
Ground truth